Lecture 4

Measurement and data curation: precision is not validity

AI-assisted reconstruction from course slides, not a lecture transcript. Discussion reasoning is inferred; classroom responses were not supplied.

The lesson. More precise estimates of a recorded quantity do not establish that the quantity represents the thing we care about. The lecture connects sampling error and model complexity to construct validity, biased observation, feedback under optimization, and documentation that supports actual use decisions.

What more data improves, and under which assumptions

For a fixed classifier evaluated on n independent draws from the target population, correctness is Bernoulli with parameter p:

$$\mathbb{E}[\widehat{p}] = p, \qquad \mathrm{SE}(\widehat{p}) = \sqrt{p(1-p)/n}.$$

Halving the standard error requires four times as many examples. At $p = .8$, $n = 25, 100, 400$ give SEs $.08, .04, .02$. This is an average precision statement: an individual added sample can move the estimate farther from the true value.

For a fixed, finite hypothesis class H, i.i.d. examples, and 0–1 loss, with probability at least $1 - \delta$,

$$\sup_{h \in H} |R(h) - \widehat{R}(h)| \leq \sqrt{\frac{\log(2|H|/\delta)}{2n}}.$$

More candidate models enlarge the bound; more data shrink it. The simultaneous guarantee differs from estimating one model fixed in advance. Regularization controls complexity, but this particular finite-class bound cannot be applied literally to an unrestricted neural-network parameter space. Optimizing a smooth surrogate such as cross-entropy is another distinction: the training objective and the reported accuracy are different quantities.

Observation and experimental design change the inference

The historical examples show data collection as purposeful work for understanding populations and acting on social conditions.

The returning-aircraft illustration is **selection/survivorship bias**: damage is observed conditional on returning. It misses aircraft that did not survive. In the fertilizer example, soil quality causes both fertilizer use and yield, so the observed association mixes the treatment effect with a **confounder**. Increasing the same biased observational sample does not by itself resolve either problem.

Blocking agricultural plots by soil conditions puts every treatment within comparable conditions; randomizing treatment positions within blocks reduces systematic assignment differences. This explains why design matters before estimation and why a correlation does not identify the desired causal effect.

Separate construct, measurement, and objective

In the health example, *need* is the construct, recorded spending is the measurement used as a target, and prediction loss is the optimized objective. Spending also reflects prices, access, and service use. Better cost prediction can therefore sharpen a poor proxy for unmet need.

Construct validity concerns whether evidence and theory support interpreting a measurement as the intended construct for a particular use. Reliability concerns consistency. A timed coding test may consistently rank speed while omitting debugging and collaboration; accurate spending records can omit unmet care. More data or repeated measurements cannot alone establish validity.

Biased labels can change the fairness conclusion

The COMPAS discussion separates recorded rearrest Y from reoffending Y^*. Detection and enforcement sit between them. If that observation process differs across groups, fairness measured against Y need not match fairness against Y^*. Varying assumptions about unrecorded offenses in a sensitivity analysis tests whether the conclusion changes; it does not reveal the unobserved truth.

Predictive policing makes a feedback mechanism visible: recorded arrests reflect police activity as well as underlying drug use. Sending officers where arrests were recorded can produce further records there. The target can partly measure the institution's past actions. The healthcare case similarly shows why equal treatment conditional on a recorded score can coexist with unequal unmet need.

Optimization can break a proxy's usefulness

Goodhart's law warns that a statistical relationship can change when a measure becomes a policy target. People or agents adapt to the incentive. In the boat-race example, points are awarded for hitting targets; the agent repeatedly circles through respawning targets instead of finishing the race. Reward rises while the intended outcome fails: *reward hacking*.

Identify the exploitable gap between measure and goal, then test a redesign under optimization. More examples of high reward do not establish success. The same reasoning applies to benchmarks repeatedly used for model selection.

Documentation is evidence for a use decision

Data pass through collection, annotation, training, and deployment. Assumptions about population, label meaning, missing values, and collection conditions can be lost at these handoffs, producing downstream data cascades. The component datasheet analogy asks engineers to specify operating conditions and limitations, as they would for a sensor.

Datasheets describe dataset purpose, composition, collection, uses, and maintenance. **Model cards** describe model uses, evaluation, and limitations. **System cards** address a system's capabilities, risks, and mitigations. They should help a reader assess suitability for a specified use.

The smile-detector example proposes selecting photographs of children. Evaluation on CelebA by annotated age and gender does not establish performance for children. A useful revision requests relevant evaluation images, label definitions, and errors by age and capture conditions. Missing evidence establishes uncertainty, not an invented failure result. This use–evidence–gap–revision chain is central to A2.

Reproduction answers a further, distinct question

Code, provenance, evaluation procedures, and reporting checklists support reproduction. Different seeds can change results with fixed algorithms and hyperparameters. For m independent runs, $s/\sqrt{m}$ estimates the mean's SE; it differs from individual-run SD and test-example uncertainty. Reproducing a score does not establish validity or deployment usefulness.

Midterm emphasis (inference). Know the fourfold-sample calculation and the interpretation and assumptions of the complexity bound. Strong conceptual priorities are validity versus reliability, rearrest versus reoffending, confounding versus selection, reward hacking, and a concrete documentation improvement with the evidence it needs. These remain in scope despite not appearing on the reference sheet.

Slide trail. L04.pdf, pp. 2–6 (estimation/complexity), 10–20 (design, validity, labels, Goodhart), 21–29 (documentation/reproduction); annotated copy consulted. A2: pp. 1–2. Page numbers refer to PDF pages, not the slides' continuous printed numbering.